Unified Bharat: Cross-Sector District Level Policy Analytics Lakehouse

INFO 584 Project Proposal – Team R2K

Overview

Different Ministries in India produce publicly available policy datasets and store them in official silos. District-level Corporate Social Responsibility (CSR) spending, groundwater quality indicators and sanitation metrics are some metrics that while conceptually linked, are reported under inconsistent schemas or naming conventions. This makes it difficult to reliably integrate due to variations in district spellings (for reasons like linguistic differences, abbreviations etc.) with standard SQL joins.

This project thus proposes to build a **distributed Lakehouse architecture** able to integrate cross ministry datasets at the district-year level and pursue the following research question:

Does subsequent improvement in groundwater quality and sanitation outcomes associate with district-level CSR spending?

The project aims to conduct a rigorous panel analysis to evaluate associations while handling district and year fixed effects.

Data Sources

The datasets to be used are public and available from the National Data and Analytics Platform (NDAP) of India.

- District-wise CSR spending (Ministry of Corporate Affairs)
- Groundwater quality indicators (Ministry of Jal Shakti)
- Swachh Bharat Mission (Gramin) sanitation metrics
- Local Government Directory (LGD) master district codes (canonical reference)

Architecture

The system will be built with Apache Spark that follows a medallion architecture.

- Raw Data: Dataset stored in distributed format

- Standardized and Integrated Data:
  - Normalization of district names (case, punctuations, abbreviation)
  - LGD codes map directly using a broadcast canonical dictionary
  - Rapid rejection of non-matching strings with bloom filter.
  - Applying candidate blocking and Levenstein distance on residual unmatched cases.

- District-year panel dataset (with derived indicators like CSR per capita, sanitation coverage rate, groundwater contamination metrics) will be stored in Delta Lake format for MERGE operations with new data.

Analytics and Methods

- Exploratory Analysis

    - Temporal trends in CSR spending and sanitation coverage.
    - Spatial variation in groundwater contamination.
    - District clustering based on multi-sector indicators.
    - Correlation heatmaps and distribution diagnostics.

- Panel Regression: To strengthen rigor outside of simple correlations, the project will estimate a a standard two-way fixed effects panel regression model (Wooldridge, 2010):

    - $Outcome_{\{d,t\}} = \beta, CSR_{\{d,t-1\}} + \alpha_d + \gamma_t + \epsilon_{\{d,t\}}$
      Here,
      - α_d= district fixed effects
      - γ_t= year fixed effects
      - $CSR_{d,t-1}$= lagged CSR spending

      With this model only conditional associations are estimated while controlling for fixed effects and not claiming any causality.

- Models (proposed):
    - Linear regression
    - Random Forest
    - Gradient Boosting (XGBoost)
    - Spark MLlib distributed regression
- Evaluation:
    - RMSE
    - R^2
    - Cross-validation error
    - Feature importance comparison

Scalability

Instead of just relying on $O(n^2)$ claims, performance improvements will be observed through distributed experiments:

- Bloom Filter Analysis
    - Derive optimal bit-array size and number of hash functions.
    - Compute expected false positive rate.
    - Validate against observed matching errors.
- Performance Study
    - Pandas vs Spark runtime.
    - Measure shuffle volume and executor time.
    - Analyze runtime scaling across partitions.
    - Evaluate incremental updates via Delta Lake MERGE operations.

References

NITI Aayog. (n.d.). *National Data & Analytics Platform (NDAP)*. Government of India. https://ndap.niti.gov.in/

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.

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